

DEV201

AWS SUMMIT



DEV201

From Prompt to Production: Building Reliable AI Agents on AWS

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Agenda

- Production Reality
- AI Agent Architecture
- Reliability Challenges
- AWS Patterns
- Takeaways



“But the demo worked perfectly...”

The Demo vs Production Gap

Why AI systems that look incredible on stage break in the real world

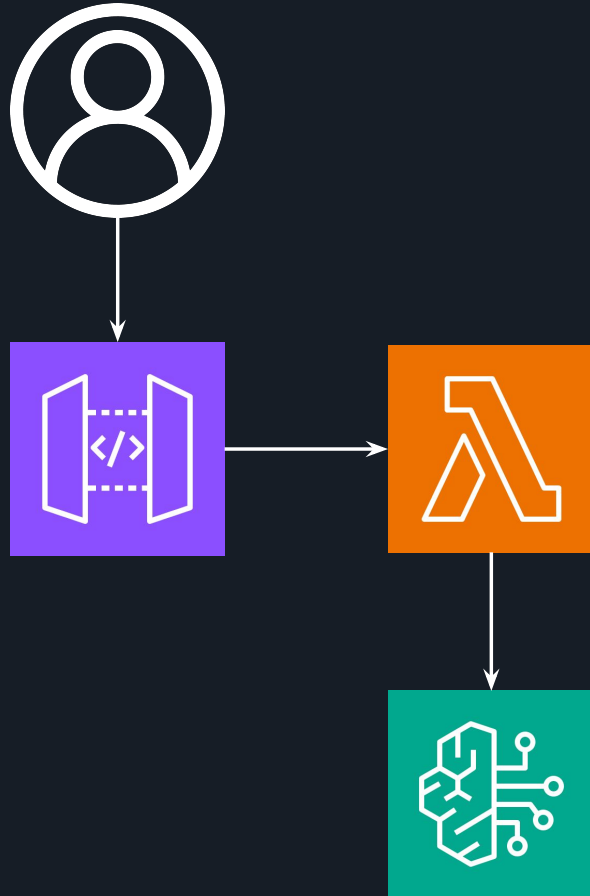
DEMO

- ✓ One happy path
- ✓ Perfect prompts
- ✓ Fast responses
- ✓ Clean outputs
- ✓ Single request

PRODUCTION

- ✗ Real users
- ✗ Ambiguous requests
- ✗ Timeouts
- ✗ Hallucinations
- ✗ Scale + concurrency

What Most People Build First



Why It Fails in Production:

No orchestration

The LLM decides everything with no guardrails

No retries

One timeout kills the entire request

No memory

Every request starts from zero

No observability

When it fails, you have no idea why

No control layer

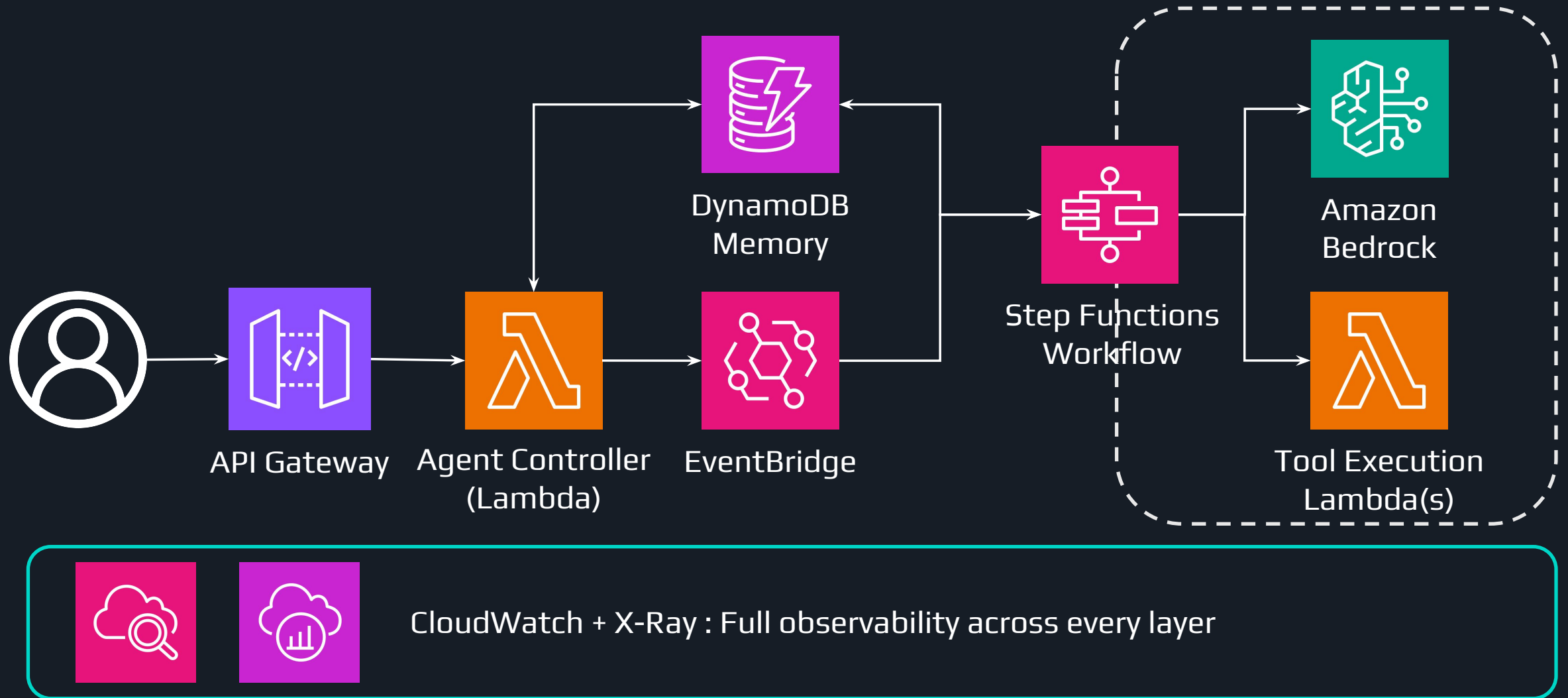
Business logic lives inside a prompt

** This is the naive solution*



"AI agents are distributed systems
wearing a chatbot costume."

Production AI Agent Architecture on AWS

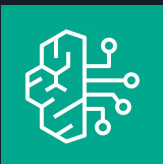


Problem #1: Hallucinations

LLMs confidently generate incorrect information. Give them verified facts instead.

RAG: Retrieval-augmented generation

```
// Retrieve before calling the model
const kb = await bedrock.retrieve({
  knowledgeBaseId: KB_ID,
  retrievalQuery: {
    text: userQuestion
  }
});
const context = kb.retrievalResults
  .map(r => r.content.text).join('\n');
```

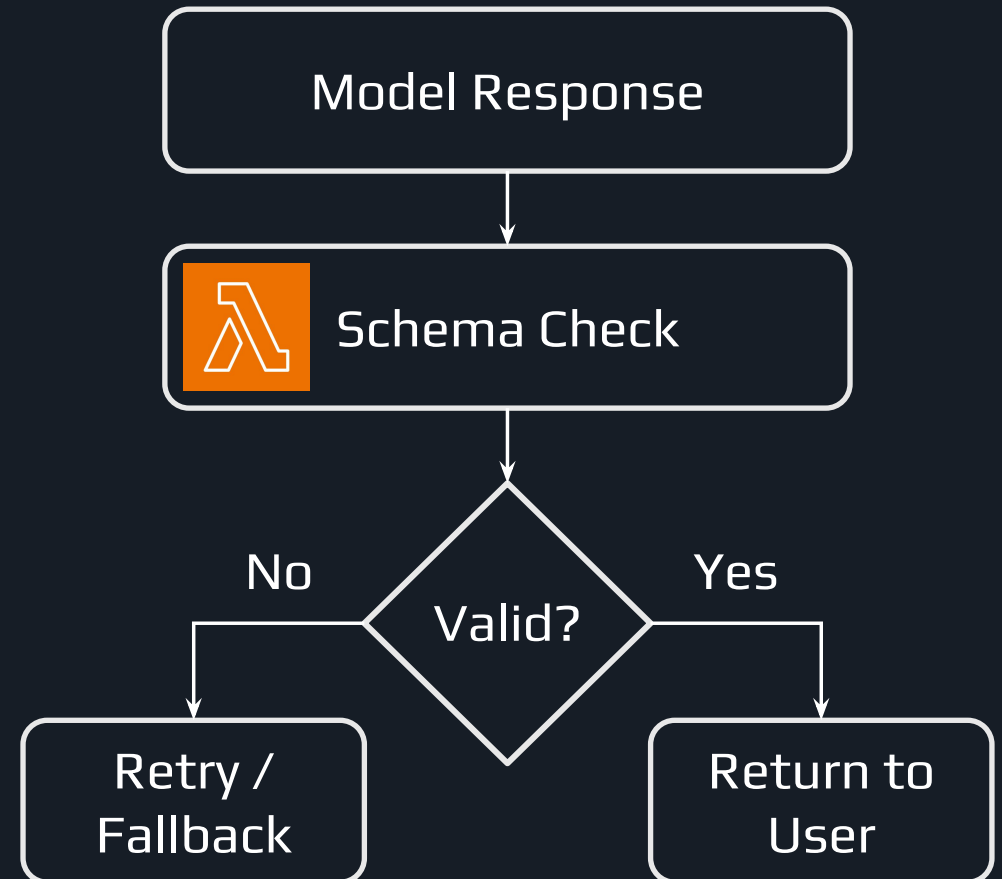


Bedrock Knowledge Bases Retrieval



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Output Validation Layer



Problem #2: Tool Execution Failures

APIs timeout, arguments malform, retries duplicate. You have to build for it explicitly

```
// Step Functions retry + DLQ
"Retry": [{
  "ErrorEquals": ["States.ALL"],
  "IntervalSeconds": 2,
  "MaxAttempts": 3,
  "BackoffRate": 2,
  "JitterStrategy": "FULL"
}]
```

Step Functions retry configuration

```
// Idempotency key before executing
const key =
  `tool:${toolName}:${hash(args)}`
;
const prev = await dynamo.get(key);
if (prev) return prev.result;

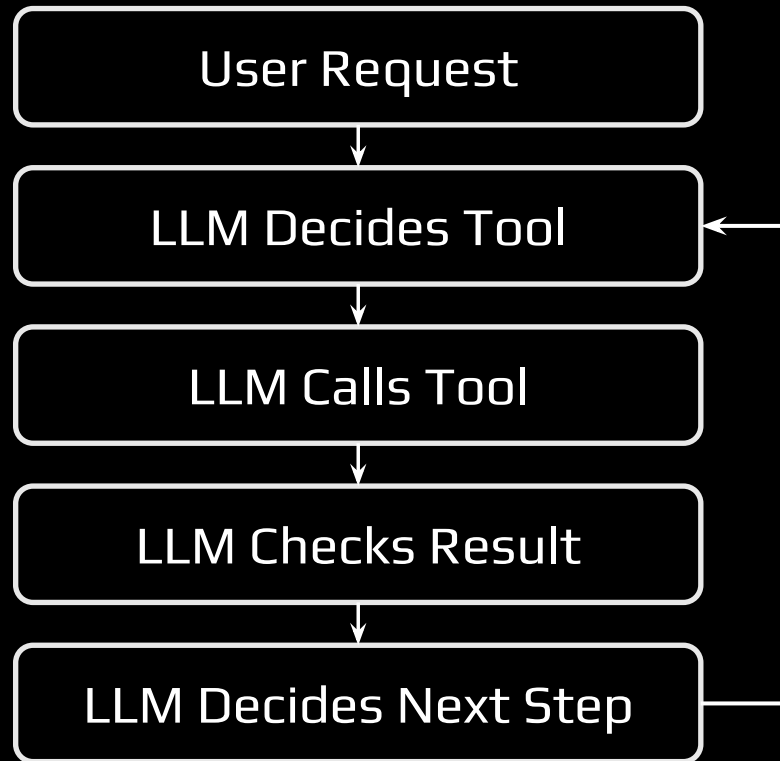
const result = await
executeTool(args);
await dynamo.put(key, result,
  { ttl: 3600 });
return result;
```

Idempotency keys for DynamoDB

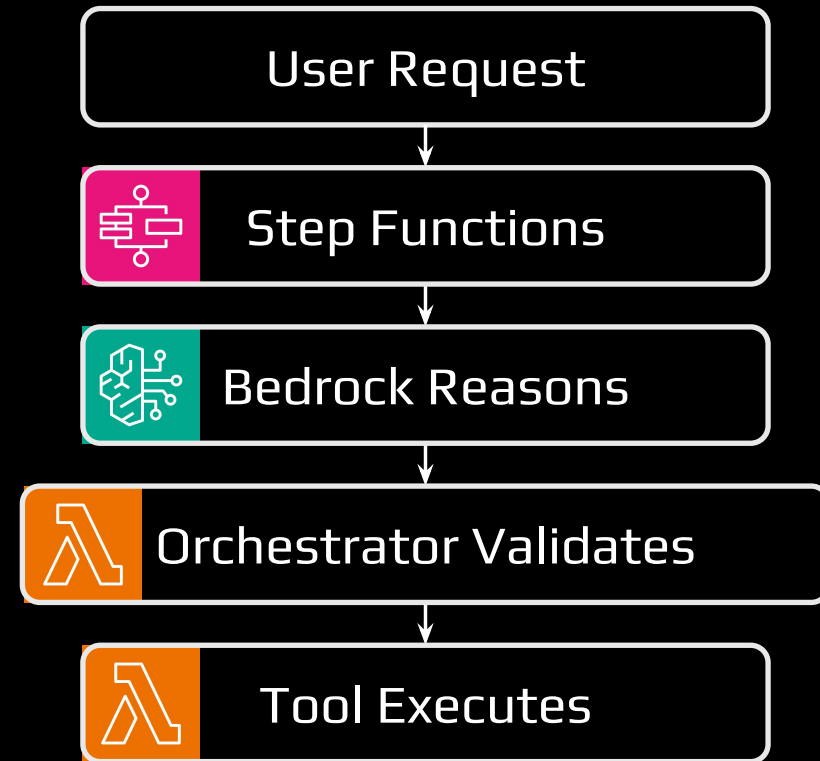
Problem #3: Orchestration Chaos

"Let the agent figure it out" causes loops, runaway costs, and impossible debugging

✗ Anti-Pattern: Agent Controls Everything



✓ Correct: System Controls Execution



Problem #4: No Observability

Traditional debugging fails for AI. You need structured signals from the start

```
// Structured log
console.log(JSON.stringify({
  requestId,
  toolName,
  inputTokens: usage.inputTokens,
  outputTokens: usage.outputTokens,
  latencyMs: Date.now() - start,
  retryCount,
  workflowPath,
  success: true
}));
```

*One log line per request, queryable in CloudWatch
Logs Insights*

```
// CloudWatch alarm
new cloudwatch.Alarm(this,
  'HighLatency', {
    metric: fn.metricDuration({
      statistic: 'p99',
      period: Duration.minutes(5),
    }),
    threshold: 10000, // 10s
    evaluationPeriods: 2,
    alarmAction: snsAlert,
  });
```

CDK alarm on p99 latency. Fires before users notice



Problem #5: State and Memory

Stateless prompts create fragile, frustrating experiences. Externalize everything

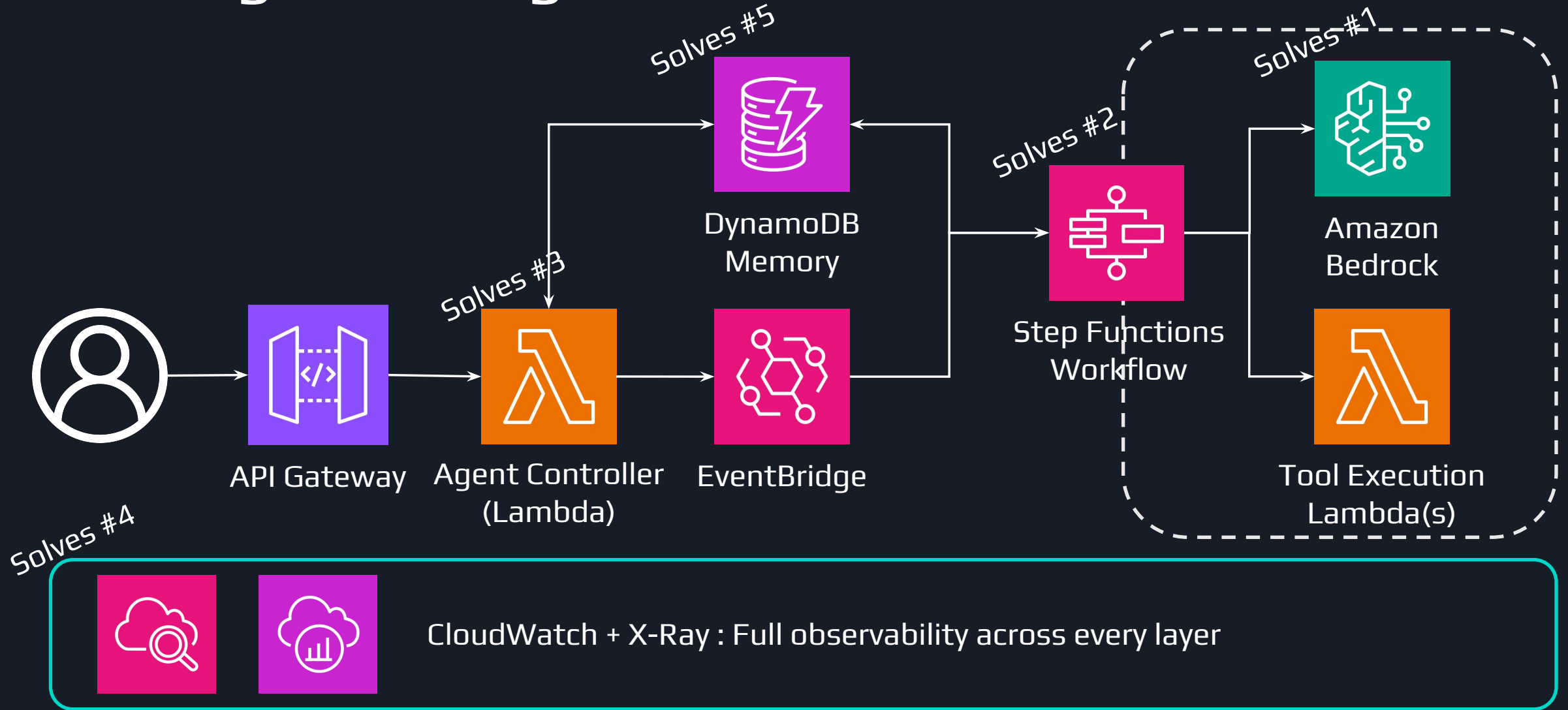
```
// Retrieve session state + assemble
const session = await dynamo.get({
  TableName: 'agent-sessions',
  Key: { sessionId }
});
const prompt = buildPrompt({
  systemContext,
  longTermMemory: session.userFacts,
  recentTurns: session.history.slice(-6),
  userMessage,
});
```

Context assembly: long-term memory + last 6 turns + new message

```
// Summarize + persist after each turn
if (session.history.length > 20) {
  session.summary = await summarize(
    session.history
  );
  session.history = [];
}
session.history.push({ role, content });
await dynamo.put('agent-sessions',
session);
```

Auto-summarize after 20 turns to stay within context limits

Putting It All Together



Demo



Key Takeaways

1

AI agents are distributed systems

Apply every pattern you already know

2

Move control out of the model

LLM reasons. Step Functions and Lambda control.

3

Design for failure, not perfection

Retries, DLOs, timeouts, idempotency keys

4

Observability is mandatory

Log prompts, tools, decisions, latency, from day one

5

The code is approachable

Everything today is Lambda + Step Functions + Bedrock



Thank you

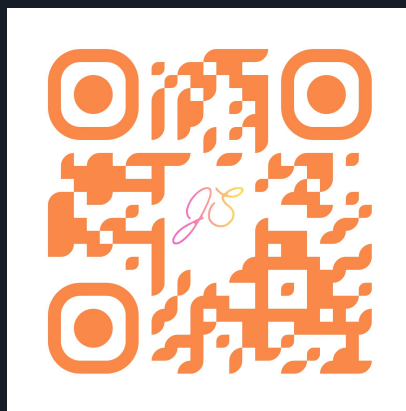
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Please complete the
session survey

◆ THE NEON SCRATCH LOUNGE

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Sandpaw

SPHINX ROGUE

[NEW GAME]

HP 80/80

XP 0/100 LVL 1

💰 25 CreditChips ↑ Neon Scratch Lounge

STR:4 AGI:9 ARC:4 STL:9

ID: 478a519a-b88a-4f35-8a6a-7a0978b2da45b

The neon lights flicker through smoke-stained windows. You pad into The Neon Scratch Lounge, your Sphinx heritage making you stand out even in this den of misfits. Madame Fluffington, a cloud of white Persian fur draped across the bar, opens one ancient eye. "Another rogue seeking fortune in Neo-Pawsburg's underbelly," she purrs, her voice like gravel wrapped in silk. "How refreshingly predictable." Glitch, a scruffy tabby with cybernetic whiskers, waves you over from a corner booth. His implants spark nervously. "New face means new opportunity. Got a job if you're interested — simple data extraction from a RoombaCore patrol unit. Pays 50 CreditChips. Just gotta get past the security in the Alley District." The jazz quartet in the corner begins a mournful saxophone number. A rat scurries past your paws. HP: 80/80 | Location: NeonScratchLounge

◆ AWS MECHANICS

WORKFLOW EXECUTION

[turn--]

✓ RetrieveLore	[Bedrock KB]
✓ InvokeDungeonMaster	[Claude 4.5]
✓ ValidateAndRoute	[Lambda]
✓ ExecuteTools	[Lambda × 3]
✓ PersistCampaign	[DynamoDB]
✓ FormatResponse	[Lambda]

CLOUDWATCH LOGS

Awaiting Lambda invocations...

METRIC	INPUT	OUTPUT	TOTAL
Tokens/turn	841	459	1300
Session total	841	459	1300

TOOL CALLS/TURN update-quest-log

RETRY COUNT 0 DLQ DEPTH 0





Sandpaw

SPHINX ROGUE

[NEW GAME]

HP

80/80

XP

0/100 LVL 1

25 CreditChips

Chrome Alley

STR:4 AGI:9 ARC:4 STL:9

ID: 6784312a-5528-4f15-a0c5-1a0790c0a45b

DICE ROLLS (1)

hide

DICE ROLL: INITIATIVE-ROLL-COMBAT-START

1

d6

1 = 1 X CRITICAL MISS

The neon lights flicker through smoke-stained windows. You pad into The Neon Scratch Lounge, your Sphinx heritage making you stand out even in this den of misfits. Madame Fluffington, a cloud of white Persian fur draped across the bar, opens one ancient eye. "Another rogue seeking fortune in Neo-Pawsburg's underbelly," she purrs, her voice like gravel wrapped in silk. "How refreshingly predictable." Glitch, a scruffy tabby with cybernetic whiskers, waves you over from a corner booth. His implants spark nervously. "New face means new opportunity. Got a job if you're interested - simple data extraction from a RoombaCore patrol unit. Pays 50 CreditChips. Just gotta get past the security in the Alley District." The jazz quartet in the corner begins a mournful saxophone number. A rat scurries past your paws. HP: 80/80 | Location: NeonScratchLounge

You bolt toward the exit with characteristic Sphinx speed. Madame Fluffington doesn't even look up-she's seen worse impulse decisions. The rain-slicked street outside ChromeAlley greets you with its usual ambiance: flickering neon, the hum of distant machinery, and yes-a MkII RoombaCore patrol drone making its rounds. Its red optical sensor sweeps the alley. It hasn't detected you yet, but you're charging straight at it. The drone's combat protocols activate. Its cleaning attachments retract, replaced by defensive tasers. Roll for initiative, then we see if those claws of yours can dent corporate property. This is either very brave or very stupid. Possibly both.

AWS MECHANICS

WORKFLOW EXECUTION

[turn--]

RetrieveLore

[Bedrock KB]

InvokeDungeonMaster

[Claude 4.5]

ValidateAndRoute

[Lambda]

ExecuteTools

[Lambda x 3]

PersistCampaign

[DynamoDB]

FormatResponse

[Lambda]

CLOUDWATCH LOGS

4 events

20:48:22.501

controller

✓

dungeon-controller-start

20:48:32.974

invoke-dm

9208ms ✓

in:841 out:334

20:48:34.600

invoke-dm

10732ms ✓

in:841 out:459

20:48:35.492

exec-tool

✓

update-quest-log

METRIC

INPUT

OUTPUT

TOTAL

Tokens/turn

2574

429

3003

Session total

3415

888

4303

TOOL CALLS/TURN

roll-dice

RETRY COUNT 0

DLQ DEPTH 0



Sandpaw

SPHINX ROGUE

[NEW GAME]

HP

80/80

XP

0/100 LVL 1

25 CreditChips

Chrome Alley

STR:4 AGI:9 ARC:4 STL:9

ID: 678a517e-584b-4f33-ba26-2a098b0d4x02

The neon lights flicker through smoke-stained windows. You pad into The Neon Scratch Lounge, your Sphinx heritage making you stand out even in this den of misfits. Madame Fluffington, a cloud of white Persian fur draped across the bar, opens one ancient eye. "Another rogue seeking fortune in Neo-Pawsburg's underbelly," she purrs, her voice like gravel wrapped in silk. "How refreshingly predictable." Glitch, a scruffy tabby with cybernetic whiskers, waves you over from a corner booth. His implants spark nervously. "New face means new opportunity. Got a job if you're interested - simple data extraction from a RoombaCore patrol unit. Pays 50 CreditChips. Just gotta get past the security in the Alley District." The jazz quartet in the corner begins a mournful saxophone number. A rat scurries past your paws. HP: 80/80 | Location: NeonScratchLounge

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You explode from the bar into ChromeAlley's neon-soaked rain. A MkII RoombaCore patrol drone rotates its sensor array toward you-red light sweeping, combat protocols engaging with mechanical precision. The drone's chassis gleams. Your claws extend. Initiative determines who strikes first. The drone calculates threat probability. You are already moving. Roll initiative (d20 + agility). Then we see if Sphinx stealth translates to Sphinx violence.

◆ AWS MECHANICS

WORKFLOW EXECUTION

[turn--]

✓ RetrieveLore

[Bedrock KB]

✓ InvokeDungeonMaster

[Claude 4.5]

✓ ValidateAndRoute

[Lambda]

✓ ExecuteTools

[Lambda x 3]

✓ PersistCampaign

[DynamoDB]

✓ FormatResponse

[Lambda]

CLOUDWATCH LOGS

5 events

20:48:22.501

controller

✓ **dungeon-controller-start**

20:48:32.974

invoke-dm

9208ms ✓ in:841 out:334

20:48:34.600

invoke-dm

10732ms ✓ in:841 out:459

20:48:35.492

exec-tool

✓ **update-quest-log**

20:50:48.012

invoke-dm

7804ms ✓ in:2574 out:416

METRIC	INPUT	OUTPUT	TOTAL
Tokens/turn	3782	406	4188
Session total	7197	1294	8491

RETRY COUNT 1

DLQ DEPTH 1 ⚠

Active Campaigns (Last Hour)

9

■ Player Actions

DLQ Depth

18

■ DLQ Messages

Monsters Defeated (Last Hour)

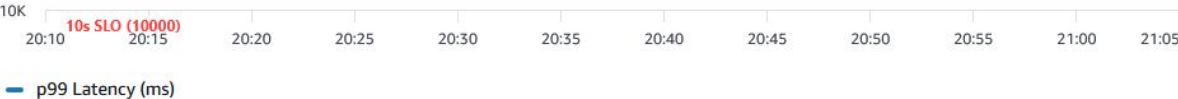
2

■ Monsters

p99 Controller Latency (target: <10s) ✎

📄 ⚙️ 🔄 📱 ⋮

No data available.
Try adjusting the dashboard time range.



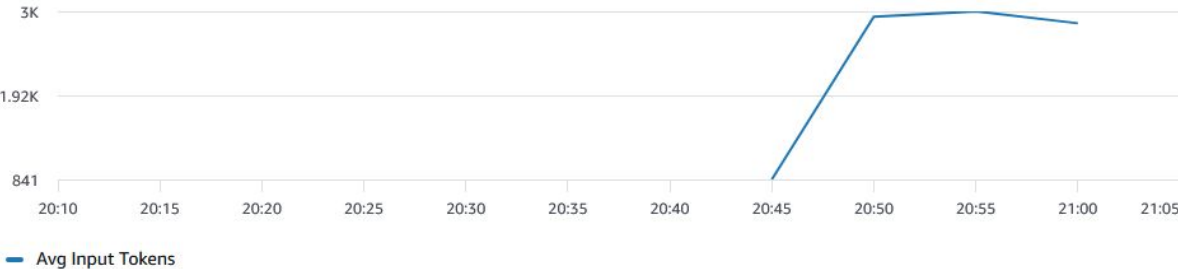
p99 DM Invocation Latency (target: <28s)

⋮



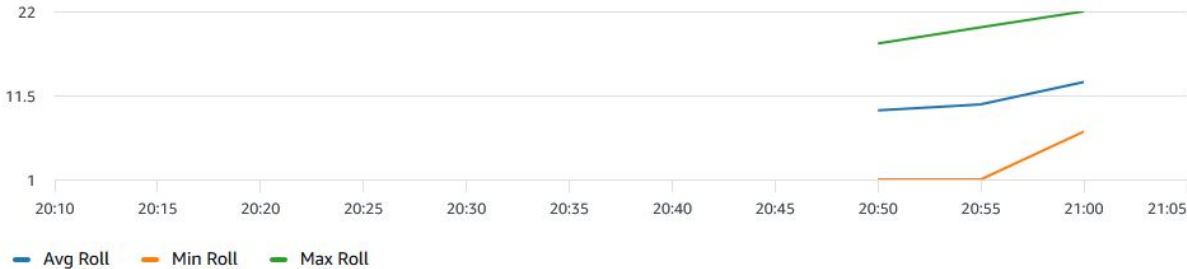
Bedrock Token Usage Per Turn

⋮



Dice Roll Distribution (d20 totals — verify fairness)

⋮



Workflow Executions



Monsters Defeated Per Hour

⋮

